PART 1 · MODULE E

Python and statistics for analysis

Pandas summaries (tested) and the statistics you must explain.

E1. Python summary (tested)

PYTHON

```
import pandas as pd
orders = pd.DataFrame({
    "channel": ["search","social","email","search","social"],
    "amount":  [1200, 800, 900, 2500, 1100],
})
summary = orders.groupby("channel")["amount"].agg(["count","sum","mean"]).round(1)
print(summary)
assert summary.loc["search","sum"] == 3700
assert summary.loc["social","count"] == 2
print("pandas summary tests passed")
```

E2. Descriptive statistics and probability

Mean, median and spread describe a typical value and how much values vary. Use the median for skewed amounts (a few large orders pull the mean up).

E3. Hypothesis testing

A test asks whether a difference could be chance. Set the question, the threshold (for example 5%), run the test, and report the result in plain words with the sample size.

VERIFICATION NOTE

Tool features, plans and prices change. Excel and Power BI features differ by version; check the official help pages. Links were not checked live in this draft.

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